

A Venkata Vinesh Kumar Reddy

AI/ML Engineer | Software Developer | RL & Intelligent Systems Enthusiast

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EDUCATION

Mahindra University, Hyderabad

B.Tech in Computer Science & Engineering | CGPA: 7.96 / 10.00

Expected: May 2027

Hyderabad, India

Coursework: Design & Analysis of Algorithms, Probability & Statistics, Optimization Techniques, Machine Learning, Deep Learning, Reinforcement Learning, Financial Mathematics, Data Structures

TECHNICAL SKILLS

Programming Languages: Python (Primary), JavaScript, SQL, C++ (DSA & Systems fundamentals)

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Reinforcement Learning, LSTM/GRU, Time Series Forecasting

Software Engineering: React, FastAPI, Docker, REST APIs, Git, Data Structures, Design & Analysis of Algorithms

Mathematics & Analytics: Probability Theory, Statistics, Optimization Techniques, Financial Mathematics, Data Analysis

TECHNICAL PROJECTS

Asset Price Prediction Platform

Python, Scikit-learn, Pandas, FastAPI, Docker

- **Problem:** Aggregating and clean-scaling historical market datasets to run predictive machine learning regression models without system latency bottlenecks.
- **Architecture & Dev:** Built a predictive model in Python using Scikit-learn (Random Forest and Gradient Boosting Regressors) to map asset indicators. Designed a FastAPI backend to expose prediction endpoints.
- **Outcome:** Achieved 92.4% price-direction prediction accuracy on historical test indicators with API response latencies under 15ms.

Weather Time-Series Forecasting Model

Python, PyTorch, NumPy, Matplotlib, Git

- **Problem:** Capture long-range sequence dependencies on multivariate meteorological sensors for weather forecasts.
- **Architecture & Dev:** Structured a deep learning time-series sequence model in PyTorch. Implemented stacked Long Short-Term Memory (LSTM) layers with customized recurrent cell states to prevent gradient degradation.
- **Outcome:** Reduced Mean Absolute Error (MAE) by 14.8% compared to traditional statistical baseline forecasts (ARIMA/SARIMAX).

Veltrix: Algorithmic Trading Dashboard & Backtester

Next.js, TypeScript, Python, FastAPI, Docker, PostgreSQL

- **Problem:** Aggregating historical stock price series and computing asset allocations using Markowitz/Sharpe optimization solvers without introducing interface rendering lag.
- **Architecture & Dev:** Programmed a Next.js dashboard using Recharts to plot price history and expected portfolio variance. Integrated a FastAPI (Python) backend using SciPy solvers to compute optimal portfolio weights.
- **Outcome:** Containerized and deployed the full-stack system using Docker, providing responsive visualization of strategy backtests and mathematical frontiers.

LEADERSHIP & EXTRACURRICULARS

Operations Lead

TEDx Mahindra University | 2024

- Coordinated operational setup and venue logistics for a flagship university event hosting 10+ speakers and 500+ attendees.

Logistics Head

Mahindra University Math Club | 2023

- 2024

- Managed logistical support, venue allocations, and advertising for math modeling competitions, attracting over 300+ students.

Competitive Chess & Strategic Problem Solving

Bangkok Open | Teplice Open | 2024

- Active chess player with strong interest in analytical thinking, calculation depth, game strategy, and rapid decision-making systems under strict time constraints.